

- 7 The three minerals below are known as double carbonates, which are carbonates containing 2 metallic elements. Double carbonates decompose in two steps, with each step involving the decomposition of one of the metallic carbonates that it contains. The decomposition of the carbonates releases carbon dioxide, which helps to smother fire, thus acting as fire retardants.

mineral	chemical formula
dolomite	$\text{CaMg}(\text{CO}_3)_2$
huntite	$\text{Mg}_3\text{Ca}(\text{CO}_3)_4$
norsethite	$\text{BaMg}(\text{CO}_3)_2$

Which row correctly shows the relative effectiveness of the minerals as a fire retardant?

- | | most effective | —————→ | least effective |
|---|----------------|--------|-----------------|
| A | dolomite | | norsethite |
| B | norsethite | | huntite |
| C | norsethite | | dolomite |
| D | huntite | | norsethite |